FINTECHATHON 2026 · TOPIC E: SME FINANCE COPILOT

Tideway

Sees the cash gap coming, plans the rescue, and leaves the decision with people.

THE LAST BRAINCELL · BUET, BANGLADESH

github.com/SadManQT/tideway

Synthetic sandbox. No real money, bank, lender, supplier or customer is connected.

01 The problem

Profitable businesses still miss payroll

Customers pay late, suppliers expect payment on time, and the gap lands on payroll day. Owners see it too late, and an AI that helps must never move money on its own.

OWNER

Will I make payroll?

Needs a clear answer days ahead, not a spreadsheet.

LENDER

Which clients are at risk?

Needs early warning across a whole portfolio, with proof.

02 How Tideway works

AI as a financial participant, with hard limits

1 See

Exact daily cash projection, three scenarios, 14-day look-ahead.

2 Predict

TidePay estimates when each invoice will really be paid.

3 Propose

Optimiser finds deferrals, early-pay discounts and reminders; negotiates with counterparties.

4 Approve

People approve with a PIN under a signed, expiring mandate; executes once.

THE GOLDEN RULE

The agent can read, analyse and propose. It cannot approve or execute.

03 Our own AI

Trained here, runs on-device

96.7%	8.56 d	8	1,000
TideLM exact tool plans (from 26.7%)	TidePay error vs 10.23 heuristic	languages understood	simulated futures per forecast

- TidePay: federated-trained lateness model, evaluated on 7,427 unseen synthetic invoices.
- TideLM: LoRA adapter on Qwen3 that turns questions and payment alerts into typed plans and draft records.
- Works without any model too: built-in multilingual answers and rule-based inbox extraction.

04 Treasury hub

What a bank asks for, generated from the ledger

PLAN

13-week cash flow

Weekly view that reconciles exactly with the daily projection. CSV export.

SCORE

Health score

Working-capital score with every formula and weight shown.

PROTECT

Payment integrity

Flags duplicates, split bills, outliers and changed payees.

SHARE

Credit passport

Signed, expiring summary for lenders; any edit fails verification.

05 Portfolio radar

Early warning for banks and large buyers

60	0.971	69%	0
SMEs scored in about 1.6 s	backtest AUC (0.928 baseline)	red-tier recall (41% baseline)	tenant records read

- Ranks every client by breach probability, days to gap and main driver; red, amber, green tiers.
- Shock slider stresses the whole portfolio and shows who moves into danger.
- Backtest on 600 synthetic SMEs with independent outcomes; advisory only, never a credit decision.

06 Security by design

Permissions decide, not detection

30/30	25/25	551	0
self-test cases	attacks blocked	automated tests	money-moving AI tools

1. **Maker-checker + PIN step-up**
   Large plans need two people, one the owner.
2. **Signed, single-use mandates**
   HMAC-bound to exact actions; expire; execute once with read-back.
3. **Kill switch, circuit breaker, payee fraud blocks**
   Plus a hash-chained audit log with an independent witness.

07 Built for everyone

Usable by anyone in the company

CLARITY

Every page explains itself

Purpose, three steps and what is safe, in plain words.

LANGUAGE

Finance terms explained on tap

About 40 terms like reserve floor and DSO, plus a one-minute tour.

TEAMS

Roles and workspaces

Owner, accountant and viewer; invite by one-time code; tenants isolated.

GLOBAL

29 currencies, 8 languages

Bangladesh, China and Singapore demo businesses.

08 Roadmap after selection

Planned next, not yet built

Phase	Planned capability
Pilot	Consented pilots with real SMEs; read-only bank-statement and accounting imports.
Bank integration	Open-banking connectors and lender API for the portfolio radar and credit passport.
Trust	Asymmetric passport signatures, independent penetration test, hosted multi-tenant deployment.
Scale	Real federated training across institutions with secure aggregation.

THANK YOU

Tideway

An AI financial participant that proposes,
and people who decide.

THE LAST BRAINCELL · BUET

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